25/9/23

Assignment - 2

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Section - A (2x5 = 10)

Q 1. What is data dictionary ? How is it used in software Engineering ?

Ans A data dictionary is a file or a set of files that includes a database's metadata. The data dictionary hold records about other objects in the database, such as data ownership, data relationship to other objects and other data. The data ~~base~~ dictionary is an essential component of any relational data base.

The data dictionary is used by the developers throughout the software development process. for ex - they use it to design the database schema, to write application code, and to create user documentation.

Q2. Explain the different betn. UML & XML.

UML	XML
• UML stand for unified Modeling language.	• XML stand for extensible Markup language.
• it is a visual type of language.	• it is a Text based language.
• it is used to design software system.	• it is used to store and exchange data.

Q3. which process model leads to s/w reuse ? why.

Ans ROM (Reuse - Oriented Model) that leads to s/w reuse in s/w engineering. ROM is a systematic approach to s/w development that emphasizes the reuse of existing software components.

ROM is a particularly good fit for projects

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where there is a large amount of existing software that can be reused. it is also a good fit for projects where there is a need to reduce development time and cost.

Q4. What is meant by software prototyping?

Ans software prototyping is the process of creating a working model of a software system before developing the final system. The protype is a simplified version of the system that is used to demonstrate its functionality and to get feedback from users.

Q5. What is ERD? How is it different from DFD?

Ans. Entity-relationship diagram (ERD) is a graphical representation of entities and their relationship in a data-base. An ERD shows how data is organized into entities, attributes and relationship, and what constraints and rules apply to the data.

And

Data Flow Diagram (DFD) is a graphical representation of the flow of data through a system. A DFD shows how data is transformed, moved, and stored in a system, and what external entities interact with the system.

Features	ERD	DFD
Focus	Structure of data in a database.	• Flow of data through a system.
Purpose	To ~~module~~ model the relationships between entities.	• To model the processes that transform and move data.
Symbols	Entities, attributes and relationships	• Processes, data stores, external entities, data flows.
Audience	Database designer, systems analysis.	• system analysis, software engineers.

• Section - B (3×7 = 21)

Q1. Explain the Evolutionary and incremental model. What are the advantages and disadvantages?

Ans. Evolutionary model is a combination of iterative and incremental model of software development life cycle. The evolutionary development model divides the development cycle into smaller, incremental waterfall models in which user are able to get access to the product at the end of each cycle.

The evolutionary model is software development model that emphasizes flexibility and adaptability. it is a cyclical model, with each cycle consisting of the following phase :-

(i) Requirement gathering and analysis :- The requirements of the s/w product are gathered and analyzed.

(ii) Design :- A high-level design of the software product is created.

(iii) implementation :- The s/w product is implemented one increment at a time.

(iv) Testing - Each increment is tested to ensure that it meets the requirements and is free of defects.

(v) Deployment - The increment is deployed to the production environment.

• incremental Model :-

Requirement
- Build 1 → Design & develop → Testing → implementation
- Build 2 → Design & develop → Testing → implementation
- Build 3 → Design & develop → Testing → implementation

The incremental modes has multiple module and each module those are requirement, design & develop Testing, implementing.

The incremental model is a good choice for projects with complex requirements, as it allows the project to be broken down into smaller, more manageable tasks. it is a good choice for projects where there is a need to deliver early and frequent release of the s/w product.

- Advantage of Evolutionary model -

→ in evo. model, a user gets a chance to experiment partially developed system.

→ it reduces the error because the core modules get tested throughly.

- Disadvantages of ev. model -

→ sometimes it is hard to divide the problem into several version that would be acceptable to the customer which can be incrementally implemented and delivered.

- Advantages of incremental model -

→ ~~Easy~~ Early increments can serves as prototypes.

→ lower risk of overall project failure.

→ most crucial and basic services are implemented first.

- Disadvantages of incremental model -

→ Hard to map requirement into small increments.

→ Hard to define the basic services that are shared by all subsequent increments.

Q2. What is known as SRS review? How is it conducted?

Ans. An SRS review, or Software Requirements Specification review, is a formal meeting where stakeholders and experts review the SRS document to identify any errors, omissions, or ambiguities. The SRS document is a critical document that describes the requirements for a software product, and it is important to ensure that it is accurate and complete.

The SRS review is typically conducted by a team of people, including:

- Business analysts :- The individuals are responsible for gathering and understanding the business requirements for the S/W Product.
- S/W engineers - These individuals are responsible for designing and implementing the S/W product.
- Quality assurance engineers :- These individuals are responsible for testing the software product to ensure that it meets the requirements.
- Other stakeholders - This may include project managers, product owners and end users.

- The SRS review is typically conducted in the following steps:

(i) Preparation - The SRS document is distributed to all participants in advance of the review.

(ii) Review - The participants review in SRS document and identify any errors, ommissions.

(iii) Discussion - The participants discuss the findings of the review and identify any areas that need to

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be addressed.

(iv) Action items - Action items are assigned to the appropriate team members to address the finding of the review.

(v) Follow-up :- The team members meet to review the action items and ensure that they have been addressed.

- Benefits of SRS review -

→ improved quality of the SRS document.
→ Reduced risk of project failure.
→ improved communication and collaboration
→ increased user satisfaction.

- Tips for conducting an effective SRS review :-

→ involve the right people.
→ Prepare for the review.
→ Facilitate the discussion.
→ Document the result.

Q3. Explain about Rapid Prototyping Techniques.

Ans. Rapid prototyping is a group of techniques used to quickly create a physical model or ~~protype~~ prototype of a product using a variety of manufacturing technologies. Rapid prototyping is often used in product design and development to test form, fit, and function before investing in the production of large quantities of parts.

- Some of the most common rapid prototyping techniques include :-

- 3D Printing :- 3D printing is a process of building a three-dimensional object from a digital model by adding material layer by layer. There are a variety of different 3D printing technologies, each with its own advantages and disadvantages.

- <u>CNC machining</u> :- CNC machining is a process of using computer-controlled machine tools to remove material from a solid block to create a desired shape. CNC machining is a very versatile process that can be used to create a wide variety of parts, from simple to complex.
- <u>Vacuum Casting</u> :- Vacuum casting is a process of creating a mold of part using silicone or other material. The mold is then filled with a liquid resin, which is cured to create a solid part. Vacuum casting is a good choice for creating prototypes of parts with complex geometries.

→ Rapid prototyping techniques can be used to create prototypes of a wide variety of products, including

- <u>Consumer products</u> :- Rapid prototyping is often used in the development of new consumer products, such as smartphones, laptops, and other electronic devices.
- <u>Automotive parts</u> :- Rapid prototyping is used to create prototypes of automotive parts, such as car bumpers, headlights, and interior trim.
- <u>Medical devices</u> :- such as surgical implants and prosthetics.
- <u>Aerospace parts</u> :- It is used in aerospace parts, such as aircraft components and rocket engines.

• Section - C (3×11 = 33)

Q1. Suppose you have to build a product to determine the inverse of 3.546784 to four decimal places. Once the product has been implemented and tested, it will be thrown away. Which life cycle model would you use? Give reasons for your answers.

Ans. I would use the throw-away prototyping model to build a product to determine to inverse of 3.546784 to four decimal places. This model is well-suited for projects with specific requirements, such as in this case, where the product is simple and has a clear goal.

The throw-away prototyping model is a software development model in which a prototype is quickly built and tested to get feedback from users and stakeholders. The prototype is then discarded, and the final product is developed based on the feedback.

The throw-away prototyping model is good choice for this project because it is -

• Fast and efficient :- The throw-away prototyping model is a fast and efficient way to develop a product for a simple task.

• Flexible :- The throw-away prototyping model is flexible, allowing for changes to the requirements and scope of the project based on user feedback.

• Low risk :- The throw-away prototyping model is a low-risk approach, as the prototype is discarded if it does not meet the needs of the users.

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Here, is a brief overview of the steps involved in the throw - away prototyping model-

(i) Requirement gathering and analysis : The first step is to gather and analyze the requirements for the product. in this case, the requirement is to develop a product to determine the inverse of 3.546784 to four decimal places.

(ii) ~~Protype~~ Prototype :- A prototype is then designed and developed based on the requirements. The prototype should be simple and easy to use, and it should focus on the core functionality of the product. in this case, the prototype would be a simple calculator that can perform the inverse operation.

(iii) Testing :- The prototype is then tasted to ensure that it meets the requirements and that it is easy to use. User feedback is also collected to identify any areas where the prototype can be improved.

(iv) Final product development :- The final product is then developed based on the feedback from the prototyping phase.

(V) Deployment :- The final product is then deployed to the production environment.

in the case of this project, the throw-away pratyping model would be used to develop a prototype calculator that can perform the inverse operation. The prototype would be tested to ensure that it can calculate the inverse of 3.546784 to four decimal places

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accurately. Once the prototype is tested and approved, the final product would be developed. The final product would be a more polished and robust version of the ~~protype~~ prototype, and it would be deployed to the production environment.

Q2. How can design reviews improve designs? How they can help to reducate operators, users, clients, analysts, engineers, coders and testers? How can design reviews uncover deficiencies in SRS?

Ans. Design reviews ~~can improve~~ can improve designs in a number of ways like :-

- Identify and correct errors and omissions : Design reviews can help to identify and correct error and omissions in the design. This can be done by having multiple people review the design and by using a variety of review techniques, such as checklists, inspections and walk-throughs.
- improve the overall quality of the design : Design reviews can help to improve the overall quality of the design by identifying and addressing areas where the design can be improved. This can include identifying areas where the design is inefficient, difficult to implement, or difficult to maintain.
- improve communication and collaboration : Design reviews can help to improve communication and collaboration between the different members of the team. This can help to ensure that everyone is on the same page and that the design is

aligned with the goals of the projects.

Design reviews can also help to educate operators, users, clients, analysts, engineers, coders, and testers in a number of ways :-

- Operators and users :- Design reviews can help operators and users to understand the design of the system and how it will work. This can help them to be more prepared to use and maintain the system once it is deployed.
- Clients : Design reviews can help clients to understand the design of the system and to ensure that it meets their requirements. This can help to reduce the risk of misunderstanding and disputes later on in the project.
- Analysts, engineers, coders, and Testers :- Design review can help analysts, engineers, coders, and testers to understand the design of the system and to identify any potential problems with the design before it is implemented. This can help to reduce the cost and effort of testing and debugging the system.

Design reviews can uncover deficiencies in the SRS in a number of ways :

- Requirement Gaps :- Design reviews can reveal gaps or inconsistencies in the software Requirement Specification (SRS). When stakeholders examine the design, they may identify missing

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feaures, conflicting requirements, or ambiguous statements that need clarification.

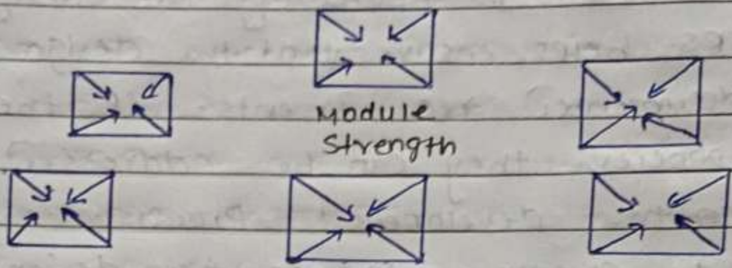
- Alignment issues :- Reviewing the design against the SRS helps ensure that the design aligns with the documented requirements. if there are discrepancies, they can be addressed and resolved early in the development process.
- Feasibility Checks - During the design review, engineers and architects can assess whether the SRS requirement are technically feasible and offer suggestions or alternatives if certain requirements are not feasible and within the given constraints.
- Risk identification :- Design reviews can uncover potential risks related to requirements. if a requirement is too complex or introduces high technical or operational risks, it can be identified and addressed before implementation.

By identifying and correcting deficiencies in the SRS, design review can help to ensure that the system is developed according to the requirements of the users and stakeholders.

Q3. Discuss Cohension and Coupling in detail.

Ans. Cohension :- The degree to which all elements of a component are directed towards a single task and all elements directed towards that task are contained in a single component.

Cohension of a unit, of a module, of an object, or a component addresses the attribu of " degree of relatedness " within that unit, module, object or component.



Cohension = Strength of relations within modules.

in computer programming, cohension defines to the degree to which the element of a module belong together. Thus cohension measures the strength of relationships betn pieces of functionality within a given module.

Types of Modules cohension as follow -

1.	Functional Cohension	Best
2.	Sequential Cohension	
3.	Communication Cohension	
4.	Procedural Cohension	
5.	Temporal Cohension	
6.	Logical Cohension	
7.	Coincidential Cohension	Worst

(i) Functional Cohension :- Functional cohension is said to exist if the different elements of a module, cooperate to achieve a single function.

(ii) Sequential Cohension :- A module is said to posses sequential cohension if the element of a module form the components of the sequence, where the output from one component of the

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Sequence is input to the next.

(iii) Communication cohension :- All the elements of the module share the same data.

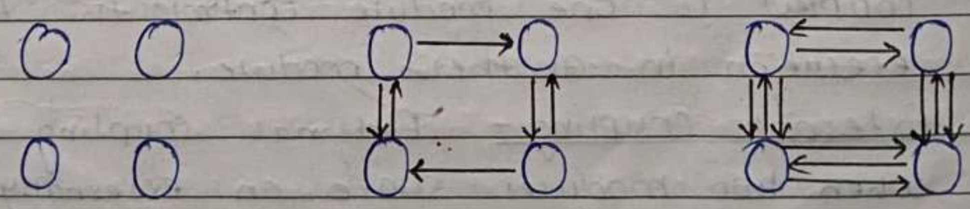
(iv) Procedural cohension :- The elements of the module are executed in a specific order.

(v) Temporal cohension - The elements of the module are executed at the same time.

(vi) Logical cohension - The elements of the module are logically related.

(vii) Coincidential cohension - The elements of the module are not logically related and are only grouped together because they are convenient to implement together.

- Coupling :- Coupling refers to the degree of interdependence between modules. A highly coupled module is one that depends on many other modules for its functionality. A low coupled module is one that is independent of other modules and does not depend on them for its functionality.



Uncoupled: no dependencies (a) | Loosely coupled: some dependencies (b) | Highly coupled Many dependencies (c)

A good design is the one that has low coupling. Coupling is measured by the number of relations betn. the modules.

- Types of Module Coupling :-

1.	No Direct Coupling
2.	Data Coupling
3.	Stamp Coupling
4.	Control Coupling
5.	Common Coupling
6.	External Coupling
7.	Content Coupling

Best

↑

Worst

(I) <u>No Direct Coupling</u> :- In this case, modules are subordinates to different modules. therefore no direct coupling.

(ii) <u>Data coupling</u> :- When data of one module is passed to another module, this is called data coupling.

(iii) <u>Stamp coupling</u> :- Two modules are stamp coupled if they share a data structure.

(iv) <u>Control coupling</u> - Two modules are control coupled if one module controls the flow of execution in another module.

(v) <u>External coupling</u> - External coupling aries when two modules share an externally imposed data format, communication protocols, or device interface. This is related to comm. to external tools and devices.

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(vi) Common coupling :- Two modules are common coupled if they share information through some global data items.

(vii) Content coupling :- Content coupling exists among two modules if they share code. e.g; a branch from one module into another module.

- Some Benefits of high cohension and low coupling :-

→ improved maintainability
→ Reduce error propagation
→ increased reusability
→ improved testability

- Difference b/w cohension & coupling —

Cohension	Coupling
• Cohension is also called Intra-module Binding.	• Coupling is also called inter-module Binding.
• Cohension shows the relationship within the module.	• Coupling shows the relationship betn. modules.
• Cohension shows the module's relative functional strength.	• Coupling shows the relative independence betn. the modules.
• in cohension, the module focuses on a single thing.	in coupling, modules are linked to the other modules.